

		disc will experience maximum friction
11	C	Applying conservation of linear momentum to the collision: $mv = (m + M)v_1$ To complete the vertical circle, the speed at the top can just be zero, unlike the case for string. Applying conservation of energy to the top and bottom of the circle: $(m + M) g (2L) = \frac{1}{2} (m + M) v_1^2$ $v_1 = \sqrt{4gL}$ Therefore $v = \frac{(m + M) \sqrt{4gL}}{m}$
12	C	For a satellite that is in orbit, the gravitational force provides for the centripetal force